

Continuous-Time Hemodynamic Trajectory Modeling via Probabilistic Score-Based Diffusion

Alexander Bowari

abowari2@illinois.edu

Abstract

This paper serves as an initial unfinished release to the full project in presenting a score-based generative framework for modeling continuous-time hemodynamic trajectories derived from electrocardiogram (ECG), photoplethysmogram (PPG), and arterial blood pressure (ABP) waveforms. Later on, this math and computational model will be tested and bench marked for efficiency and accuracy, however all the work will remain constant from the framework. Building on denoising diffusion probabilistic models (DDPMs) and their variance-preserving stochastic differential equation (SDE) formulation, we derive the forward noising process, its continuous-time limit, the associated reverse-time SDE, and the denoising score-matching objective used to train a time-conditioned residual convolutional network (known informally as a U-Net) that approximates the score of the noised data distribution. We motivate a score/noise parameterization over a direct mean-variance $(\mu_\theta, \sigma_\theta)$ parameterization of the reverse transition, and show why the latter is prone to vanishing and exploding gradients across the diffusion horizon while the former is not. We further extend the model to a neural stochastic differential equation (SDE) over the physiological state space, trained via the reparameterization trick and a forward sensitivity recursion for backpropagation through time. Finally, we describe how the learned score field and trajectory likelihood can be converted into a continuous, patient-specific hemodynamic risk signal. The model is trained on continuous-time vital-sign data derived from the University of California Irvine’s ”Cuff-Less Blood Pressure Estimation” dataset. This work is intended as a methodological contribution at the intersection of generative modeling and physiological monitoring, and does not constitute a validated clinical decision-support tool.

1 Introduction

Continuous physiological monitoring produces high-frequency, high-dimensional, noisy multivariate time series (ECG, PPG, ABP) whose joint dynamics encode a patient’s hemodynamic state. Two facts make this signal difficult to use for automated risk assessment. First, the mapping from raw waveform morphology to downstream cardiovascular risk is nonlinear and patient-specific. Second, classical discriminative approaches (e.g., regression or classification on hand-engineered waveform features) collapse the trajectory to a point estimate and discard information about how unusual or improbable a given trajectory is relative to the population of physiologically plausible trajectories: it fails to include any sort of Brownian or stochastic element in the evolution of the trajectory’s state.

Generative modeling offers an alternative: rather than directly regressing a risk label, we can learn the actual probability distribution of hemodynamic trajectories and use the model’s own notion of likelihood, or the magnitude of the score function needed to denoise an observed trajectory back toward the data manifold, as a continuous measure of physiological surprise. Score-based diffusion models are a natural fit for this task: they are trained purely by denoising, so they

provide a well-defined probability-flow formulation for likelihood evaluation, something necessary for a signal as structured as ECG, PPG, and ABP because they model the full data distribution and how it might evolve over time rather than a single conditional mean.

This paper aims to make the following contributions: give a self-contained derivation of the discrete-time forward noising process for hemodynamic waveforms, its continuous-time SDE limit and the corresponding reverse-time generative SDE, derive the denoising score-matching training objective from Tweedie’s formula and explain why the model cannot directly observe $\nabla_x \log p(x)$ and must instead estimate it via a tractable conditional surrogate, document a parameterization failure mode we encountered when directly regressing the reverse transition’s mean μ_θ and variance, extend the fixed-noise-schedule diffusion model to a full neural SDE over the physiological state ($dX_t = \mu_\theta(X_t, u_t, t) dt + \sigma_\theta(X_t, u_t, t) dW_t$) trained end-to-end with a reparameterization Euler-Maruyama simulator and a forward-mode sensitivity recursion for gradient computation, and finally describe how the trained score field is converted into a continuous hemodynamic risk signal, and detail the data preparation pipeline and network architecture used in the implementation of the model.

2 Data

We use continuous-time vital-sign recordings derived from the UCI ”Cuff-less Blood Pressure Estimation” dataset, which provides simultaneously recorded, preprocessed ECG, PPG, and ABP waveforms sampled at a common rate. Each recording is treated as a multichannel trajectory

$$x_0 \in \mathbb{R}^{C \times L}, \quad C = 3 \text{ (ECG, PPG, ABP)}, \quad L = \text{window length},$$

which we segment into fixed-length windows, per-channel normalize, and treat as a single data point $x_0 \sim p_{\text{data}}$ for the diffusion model of Sections 3-5. For the trajectory-forecasting extension of Section ??, the same channels are instead treated as the observed component of a latent physiological state X_t evolving continuously in time with covariate function u_t .

3 Forward Diffusion Process

3.1 Discrete-time noising process

Diffusion models define a fixed, non-learned forward Markov chain that progressively destroys structure in the data by repeated injection of Gaussian or random noise, until the terminal distribution is indistinguishable from a regular Gaussian, or just pure noise. Given data $x_0 \sim p_{\text{data}}$, the single-step transition is

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I), \quad t = 1, \dots, T, \quad (1)$$

where $\beta_t \in (0, 1)$ is a variance schedule controlling the amount of noise injected at step t . Equation (1) can be sampled as

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \varepsilon_{t-1}, \quad \varepsilon_{t-1} \sim \mathcal{N}(0, I). \quad (2)$$

Defining $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$, Eq. (2) composes in closed form:

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I), \quad x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I). \quad (3)$$

the forward chain converges to a tractable, sampleable reference distribution. This closed form is what makes training efficient because an arbitrary noise level x_t can be sampled in a single step from x_0 without simulating the full chain.

3.2 Continuous-time (SDE) limit

To connect the discrete chain to a stochastic differential equation and relate $\bar{\alpha}_t$ to the noise schedule $\beta(t)$, let $\Delta t = 1/N$ and rescale the per-step schedule as $\tilde{\beta}_i = N\beta_i$ (a noise rate with units of inverse time). With $x_i = x(t + \Delta t)$ interpreted as the state after one Euler step, Eq. (1) becomes

$$x(t + \Delta t) = \sqrt{1 - \beta(t + \Delta t) \Delta t} x(t) + \sqrt{\beta(t + \Delta t) \Delta t} \varepsilon(t). \quad (4)$$

Let $u = \beta(t + \Delta t) \Delta t$. Since $u \rightarrow 0$ as $\Delta t \rightarrow 0$, we Taylor-expand $f(u) = \sqrt{1 - u}$ about $u = 0$:

$$f(u) = (1 - u)^{1/2}, \quad f'(u) = -\frac{1}{2}(1 - u)^{-1/2}, \quad f'(0) = -\frac{1}{2},$$

so that $f(u) \approx 1 - \frac{1}{2}u$ to first order, giving

$$\sqrt{1 - \beta(t + \Delta t) \Delta t} \approx 1 - \frac{1}{2}\beta(t + \Delta t) \Delta t.$$

Substituting into Eq. (4) and using $\beta(t + \Delta t) \approx \beta(t)$ to leading order,

$$x(t + \Delta t) - x(t) \approx -\frac{1}{2}\beta(t) x(t) \Delta t + \sqrt{\beta(t) \Delta t} \varepsilon(t). \quad (5)$$

Identifying the Brownian increment $dW = \sqrt{\Delta t} \varepsilon(t)$ and taking $\Delta t \rightarrow 0$ turns Eq. (5) into the variance-preserving SDE of Song et al. [3]:

$$\boxed{dx = -\frac{1}{2}\beta(t) x(t) dt + \sqrt{\beta(t)} dW, \quad f(x, t) = -\frac{1}{2}\beta(t)x(t), \quad g(t) = \sqrt{\beta(t)}. \quad (6)}$$

Equation (6) is the continuous-time generalization of Eq. (1): it corrupts a hemodynamic waveform with a linear drift back toward the origin plus additive white noise whose intensity is governed by $\beta(t)$, and it recovers the discrete chain exactly under Euler-Maruyama discretization with step Δt .

4 Score Estimation via Tweedie's Formula

Sampling from the reverse of Eq. (6) requires the score of the noised marginal, $\nabla_x \log p_t(x)$, which is not directly observable because we only have samples, not a closed-form density we can take the gradient of. We derive the connection between the score and the denoising problem with Tweedie's formula, then show why this motivates denoising score matching as a tractable surrogate objective.

4.1 The posterior mean minimizes mean-squared error

Consider a single forward corruption $y = x + \sigma\varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$, so $p(y | x) \propto \exp(-\frac{1}{2\sigma^2}\|y - x\|^2)$. Given a noisy observation y , the estimator $\hat{x} = \hat{x}(y)$ that minimizes expected squared error is

$$\hat{x} = \arg \min_{\hat{x}} \mathbb{E}[\|\hat{x} - x\|^2 | y]. \quad (7)$$

Expanding the squared norm as a dot product, $\|\hat{x} - x\|^2 = (\hat{x} - x)^\top (\hat{x} - x) = \|\hat{x}\|^2 - 2\hat{x}^\top x + \|x\|^2$, so

$$\mathbb{E}[\|\hat{x} - x\|^2 | y] = \|\hat{x}\|^2 - 2\hat{x}^\top \mathbb{E}[x | y] + \mathbb{E}[\|x\|^2 | y]. \quad (8)$$

Then, to get the minimum of the function, we start differentiating with respect to $\hat{x}$ and setting the result to zero (the last term does not depend on $\hat{x}$):

$$2\hat{x} - 2\mathbb{E}[x | y] = 0 \implies \boxed{\hat{x} = \mathbb{E}[x | y]}, \quad (9)$$

the minimum mean squared error (MSE) estimator is exactly the posterior mean. This is the same quantity that a diffusion model's denoiser is trained to output at each noise level.

4.2 Tweedie's formula

We now express $\nabla_y \log p(y)$ in terms of $\mathbb{E}[x | y]$. By definition, $p(y) = \int p(y | x) p(x) dx$, and since $\nabla_y \log p(y) = \nabla_y p(y) / p(y)$, we differentiate as such:

$$\nabla_y p(y) = \int \nabla_y p(y | x) p(x) dx. \quad (10)$$

For the Gaussian likelihood, $\nabla_y p(y | x) = -\frac{1}{\sigma^2}(y - x) p(y | x)$, so

$$\nabla_y p(y) = -\frac{1}{\sigma^2} \int (y - x) p(y | x) p(x) dx = -\frac{1}{\sigma^2} \left[y \underbrace{\int p(y | x) p(x) dx}_{= p(y)} - \int x p(y | x) p(x) dx \right]. \quad (11)$$

The second integral is, by Bayes' rule, $p(y) \mathbb{E}[x | y]$. Dividing through by $p(y)$:

$$\nabla_y \log p(y) = -\frac{1}{\sigma^2} (y - \mathbb{E}[x | y]) \iff \boxed{\mathbb{E}[x | y] = y + \sigma^2 \nabla_y \log p(y)}. \quad (12)$$

Equation (12) is Tweedie's formula: it states that the Bayes optimal denoiser and the score of the noisy marginal determine each other exactly. Consequently, learning to denoise is essentially just learning the score, except with the scaling term σ^2 .

4.3 From an intractable marginal score to a tractable conditional target

Equation (12) still requires $\nabla_y \log p(y)$, and $p(y) = \int p(y | x) p(x) dx$ is an intractable marginalization over the unknown data density $p(x)$. Vincent's denoising score-matching identity [4] resolves this by showing that minimizing

$$\mathbb{E}_{x_0 \sim p_{\text{data}}, x_t \sim q(x_t | x_0)} \left[\|s_\theta(x_t, t) - \nabla_{x_t} \log q(x_t | x_0)\|^2 \right] \quad (13)$$

has the same minimizer, in expectation, as matching s_θ to the true but intractable marginal score $\nabla_{x_t} \log p_t(x_t)$. Crucially, $q(x_t | x_0)$ in Eq. (13) is known in closed form, it is exactly Eq. (3), $q(x_t | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I)$. For a Gaussian $\mathcal{N}(x; \mu, \sigma^2 I)$, $\nabla_x \log \mathcal{N}(x; \mu, \sigma^2 I) = -(x - \mu) / \sigma^2$, so with $\mu = \sqrt{\bar{\alpha}_t} x_0 =: a_t x_0$ and variance $\sigma_t^2 = 1 - \bar{\alpha}_t$,

$$\nabla_{x_t} \log q(x_t | x_0) = -\frac{x_t - a_t x_0}{\sigma_t^2}. \quad (14)$$

But by Eq. (3), $x_t - a_t x_0 = \sigma_t \varepsilon$ exactly, so this simplifies to

$$\nabla_{x_t} \log q(x_t | x_0) = -\frac{\varepsilon}{\sigma_t}. \quad (15)$$

This is the key simplification: the intractable marginal score is replaced by a quantity computable directly from the (known) noise ε that was sampled to construct x_t in the first place. The training objective becomes

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \varepsilon} \left[\|s_\theta(x_t, t) + \varepsilon / \sigma_t\|^2 \right] \iff \mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \varepsilon} \left[\|\varepsilon_\theta(x_t, t) - \varepsilon\|^2 \right], \quad (16)$$

where the second, equivalent form defines a noise-prediction network $\varepsilon_\theta := -\sigma_t s_\theta$, which regresses the standard-normal noise sample directly (this is the ε -parameterization of Ho et al. [2]; see Section 6). For this model, we will use the derived equation comparing $s_\theta(x_t, t)$ to the noise ε divided by the non-linear cosine noise schedule $\beta(t)$.

5 The Reverse-Time Generative Stochastic Differential Equation

Anderson’s time-reversal theorem [5] states that the reverse of a forward SDE $dx = f(x, t) dt + g(t) dW$ (running $t : T \rightarrow 0$) is itself a diffusion, thus:

$$dx = \left[f(x, t) - g(t)^2 \nabla_x \log p_t(x) \right] dt + g(t) d\bar{W}, \quad (17)$$

where $\bar{W}$ is a Wiener process for time running backward and dt is an infinitesimal time step. Substituting the SDE coefficients from Eq. (6) and the learned score $s_\theta \approx \nabla_x \log p_t(x)$ from Section 4, we get a form in terms of our variables:

$$\boxed{dx = \left[-\frac{1}{2}\beta(t)x - \beta(t)s_\theta(x, t) \right] dt + \sqrt{\beta(t)} d\bar{W}.} \quad (18)$$

Discretizing Eq. (18) with Euler-Maruyama on a step size Δt gives the ancestral sampler used at inference time:

$$x_{t-1} = x_t + \left[-\frac{1}{2}\beta(t)x_t - \beta(t)s_\theta(x_t, t) \right] \Delta t + \sqrt{\beta(t)} \sqrt{\Delta t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad (19)$$

which is initialized at $x_T \sim \mathcal{N}(0, I)$ and iterated down to $t = 0$ to synthesize a hemodynamic waveform consistent with the learned data distribution. Because the same s_θ also parametrizes the deterministic ODE of the probability flow $dx = [f(x, t) - \frac{1}{2}g(t)^2 s_\theta(x, t)]dt$, the model additionally admits exact likelihood evaluation via the instantaneous change-of-variables formula, which we use in Section 8.

6 Avoiding Vanishing/Exploding Gradients: Score vs. Mean Variance Parameterization

An apparently natural parameterization of the reverse transition is to directly model its mean and variance through a regression, where $q_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_\theta(x_t, t)^2 I)$, and train by maximizing the evidence lower bound term-by-term. This way, you’re able to perform a regression with discrete points but form a continuous trajectory, just like diffusion would be able to. In our initial implementation this was tested, with parameterized values for μ_θ and σ_θ directly which would combine to form a Euler Maruyama discretization for the path with each little change dX , affected by a deterministic component and a random Brownian term:

$$dX_t = \mu_\theta(X_t, u_t, t) dt + \sigma_\theta(X_t, u_t, t) dW_t, \quad (20)$$

where $X_t \in \mathbb{R}^d$ is the (possibly latent) physiological state, u_t collects observed covariates, μ_θ is a learned drift, and σ_θ a learned (state-dependent) diffusion coefficient.

In practice, this method works less efficiently and reliably because of the famous vanishing and exploding gradient problem, because directly parameterizing both μ_θ and σ_θ can lead to poorly conditioned optimization and vanishing or exploding gradients. During backpropagation through a sequence of transition steps, gradients involve products of partial derivatives to update the weights during gradient descent:

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\partial X_t} \prod_{t=1}^T \frac{\partial X_t}{\partial X_{t-1}} \frac{\partial X_{t-1}}{\partial \theta} \quad (21)$$

where X_t is the current state at time t . When these derivative terms are sufficiently small, their repeated multiplication causes the gradient to shrink towards zero across the network and the

network struggles to learn. Conversely, when these derivative terms are particularly large, the resulting weight updates can become enormous and training becomes unstable. Thus, this solution to modeling a hemodynamic trajectory in continuous time is impractical, as gradients tend to decay or grow exponentially with the number of steps, respectively, which is particularly problematic when μ_θ and σ_θ are learned directly since errors in the estimated drift and diffusion coefficient propagate through the stochastic trajectory and jointly affect subsequent transition distributions.

For our case, let $g_n := \partial X_n / \partial \theta$. Differentiating Eq. (29) with respect to θ and applying the chain rule (writing $\mu = \mu(X_n, \theta)$, $\sigma = \sigma(X_n, \theta)$ for brevity),

$$\frac{d\mu}{d\theta} = \frac{\partial\mu}{\partial X_n} \frac{\partial X_n}{\partial\theta} + \frac{\partial\mu}{\partial\theta}, \quad \frac{d\sigma}{d\theta} = \frac{\partial\sigma}{\partial X_n} \frac{\partial X_n}{\partial\theta} + \frac{\partial\sigma}{\partial\theta}, \quad (22)$$

which gives the forward sensitivity recursion

$$g_{n+1} = g_n + \Delta t \left(\frac{\partial\mu_n}{\partial X_n} g_n + \frac{\partial\mu_n}{\partial\theta} \right) + \sqrt{\Delta t} \varepsilon_n \left(\frac{\partial\sigma_n}{\partial X_n} g_n + \frac{\partial\sigma_n}{\partial\theta} \right), \quad (23)$$

which we write compactly as a recursion in g_n :

$$g_{n+1} = \left(I + \Delta t \frac{\partial\mu_n}{\partial X_n} + \sqrt{\Delta t} \varepsilon_n \frac{\partial\sigma_n}{\partial X_n} \right) g_n + \Delta t \frac{\partial\mu_n}{\partial\theta} + \sqrt{\Delta t} \varepsilon_n \frac{\partial\sigma_n}{\partial\theta}. \quad (24)$$

Therefore, the diffusion formulation mitigates this optimization difficulty by training a network to predict the score (or an equivalent noise parameterization) at a specified noise level, providing a bounded, explicitly controlled training target rather than requiring direct regression of both transition moments.

7 Network Architecture

7.1 Sinusoidal time embedding

The score/noise network is conditioned on the continuous diffusion time t via a fixed sinusoidal positional encoding [8], which provides the network with a smooth, high-resolution representation of t without requiring it to be learned:

$$\omega_i = \frac{1}{10000^{2i/D}}, \quad \theta_i = t \cdot \omega_i, \quad \text{PE}(t) = [\sin \theta_0, \cos \theta_0, \sin \theta_1, \cos \theta_1, \dots] \in \mathbb{R}^D, \quad (25)$$

where i indexes the embedding dimension and D is the total embedding size. The frequencies ω_i are geometrically spaced so that $1/10000 \leq \omega_i \leq 1$, giving the embedding both fast-varying components (fine discrimination between adjacent time steps) and slow-varying components (coarse discrimination across the whole horizon). PE(t) is passed through a small multilayer perceptron (basically a feed forward layer) and injected into every residual block (Section 7.3) via a learned scale or shift transformation of that block's feature map.

7.2 1D convolutional layers

Each channel (ECG, PPG, ABP) is a 1D time series, so the network's convolutional layers operate over the temporal axis. For an input feature map $h \in \mathbb{R}^{C_{\text{in}} \times L}$ and a learned kernel $w \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times K}$ (kernel size K), the output at channel c (with output c') and position ℓ is

$$(h * w)[c, \ell] = \sum_{c'=1}^{C_{\text{in}}} \sum_{k=0}^{K-1} h[c', \ell + k - \lfloor K/2 \rfloor] w[c, c', k] + b[c], \quad (26)$$

which is essentially just a sliding weighted sum over a local temporal window of width K , summed across input channels, plus a learned bias. Stacking M such layers grows the network’s temporal field linearly as $1 + M(K - 1)$, or exponentially if dilated convolutions with dilation 2^m at layer m are used; the latter is preferred here so that the network can relate distant events without an impractically deep stack. Downsampling is performed with stride-2 convolutions (halving L but doubling C), and upsampling with transposed convolutions or nearest-neighbor upsampling followed by a stride-1 convolution, mirroring the standard 2D U-Net design [6] but with all operators reduced to one temporal dimension.

7.3 Residual blocks

Each stage of the U-Net is built from residual blocks of the form

$$h' = h + \text{Conv}_2\left(\phi\left(\text{Norm}_2(\gamma(t) * \text{Conv}_1(\phi(\text{Norm}_1(h))) + \delta(t))\right)\right), \quad (27)$$

where Norm is GroupNorm, ϕ is the SiLU nonlinearity (Sigmoid Linear Unit), and $(\gamma(t), \delta(t))$ are the scale/shift parameters produced from the time embedding of Section 7.1. The identity skip connection is what keeps gradients well-scaled through very deep stacks of these blocks, producing an architectural mitigation of the vanishing-gradient problem to the parameterization fix of Section 6.

7.4 Cosine noise schedule

Rather than the original linear schedule β_t linearly interpolated between β_1 and β_T , we use the cosine schedule of Nichol & Dhariwal [7], which specifies $\bar{\alpha}_t$ directly:

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, \quad f(t) = \cos^2\left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2}\right), \quad \beta_t = 1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}, \quad (28)$$

with a small offset s (we use $s = 0.008$) that prevents β_t from being too small near $t = 0$. The linear schedule destroys high-frequency structure almost immediately, which is undesirable for ECG/PPG signals whose diagnostic content (QRS complex morphology, dicrotic notch timing) is carried largely by fine, high-frequency structure; the cosine schedule keeps $\bar{\alpha}_t$ close to 1 for longer at small t , giving the model a more gradual, and more perceptually/physiologically uniform, corruption trajectory.

7.5 Euler-Maruyama simulation and reparameterization

Trajectories are simulated with the same discretization used in Section 5:

$$X_{n+1} = X_n + \mu_\theta(X_n, t_n) \Delta t + \sigma_\theta(X_n, t_n) \sqrt{\Delta t} \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, I), \quad (29)$$

trained against observed trajectories $\{y_n\}$ with loss $\mathcal{L} = \sum_{n=1}^N \|X_n - y_n\|^2$ (or the corresponding Gaussian negative log-likelihood if σ_θ is also supervised). Because the noise ε_n is sampled from a fixed distribution that does not depend on θ , we can apply the reparameterization trick: sample all $\varepsilon_0, \dots, \varepsilon_{N-1} \sim \mathcal{N}(0, I)$ once, up front, and treat them as constants for the duration of one training step. Under this reparameterization, X_n becomes a deterministic, differentiable function of $(\theta, X_0, \varepsilon_0, \dots, \varepsilon_{n-1})$, and the recursion in Eq. (29) can be backpropagated through directly.

8 From Score/Likelihood to Hemodynamic Risk

The generated hemodynamic trajectories can be used to estimate the probability of clinically relevant events over a given forecast horizon. Starting from the same observed physiological state, the neural stochastic differential equation can be sampled repeatedly to generate N possible future trajectories,

$$X_t^1, X_t^2, \dots, X_t^N \quad (30)$$

where each trajectory represents one possible evolution of the patient’s physiological state. These trajectories form a distribution over future states, allowing risk to be estimated from the proportion of trajectories that satisfy a specified clinical condition. For example, if the event of interest is a mean arterial pressure (referred to as MAP) below 30mmHg, the probability of this event at the end of the forecast horizon can be estimated as:

$$P(MAP < 30) \approx \frac{1}{N} \sum_{i=1}^N (MAP_t^i < 30) \quad (31)$$

This allows the model to provide a probabilistic risk estimate rather than a single deterministic prediction. The same approach can be extended to events occurring at any point during the forecast horizon, such as

$$P(\min[0, T]) = MAP < 30, \quad (32)$$

which represents the probability that MAP falls below 30mmHg at any point at all during the predicted trajectory. More generally, the joint evolution of ABP, PPG, ECG, and other physiological variables across the generated trajectories can be used to estimate the probability of a range of clinically relevant hemodynamic events, which was the original goal of the model in the initial phase of creation.

9 Conclusion

We derived, from first principles, a complete score-based diffusion pipeline for continuous-time hemodynamic waveforms: the discrete forward process and its SDE, the denoising score-matching objective via Tweedie’s formula, the reverse-time generative SDE and its Euler-Maruyama sampler, and a neural-SDE extension for full trajectory forecasting documented to struggle due to vanishing and exploding gradients under a direct μ_θ/σ_θ parameterization of the reverse transition. Finally, we outlined how the resulting score field and trajectory likelihood can be utilized to predict hemodynamic failure without additional supervision, as a continuous hemodynamic risk signal. Future work includes validating this risk signal against adverse-outcome labels, conditioning the reverse process on patient covariates, and extending the neural stochastic differential equation to jointly model interventions as a controlled diffusion process.

References

- [1] J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, S. Ganguli. *Deep Unsupervised Learning using Nonequilibrium Thermodynamics*. ICML, 2015.
- [2] J. Ho, A. Jain, P. Abbeel. *Denoising Diffusion Probabilistic Models*. NeurIPS, 2020.
- [3] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, B. Poole. *Score-Based Generative Modeling through Stochastic Differential Equations*. ICLR, 2021.

- [4] P. Vincent. *A Connection Between Score Matching and Denoising Autoencoders*. Neural Computation, 2011.
- [5] B. D. O. Anderson. *Reverse-Time Diffusion Equation Models*. Stochastic Processes and their Applications, 1982.
- [6] O. Ronneberger, P. Fischer, T. Brox. *U-Net: Convolutional Networks for Biomedical Image Segmentation*. MICCAI, 2015.
- [7] A. Nichol, P. Dhariwal. *Improved Denoising Diffusion Probabilistic Models*. ICML, 2021.
- [8] A. Vaswani et al. *Attention Is All You Need*. NeurIPS, 2017.